

Marc Berthelette

Senior Engineer · AI-Assisted Engineering · Python Platforms · DevOps

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linkedin.com/in/marc-berthelette-3aba6917 · Bilingual: French (native) · English (fluent) · Available immediately

Professional Summary

Senior engineer combining roughly seven years of production software development with deep platform, automation, and reliability experience. Builds Python services and internal platforms for regulated enterprises, including a configurable validation system for 250 financial-services artifact repositories. Recently introduced AI-assisted engineering practices with hard scope gates and human-validation checkpoints before generated code reaches production. Known for learning unfamiliar systems under a delivery deadline, expanding narrow assignments into durable platform capabilities, and treating governance and reliability as first-class engineering requirements.

Core Expertise

AI-Assisted Engineering	Guardrails for generated code, hard scope gates, human-validation checkpoints, safe promotion practices, technical risk assessment
Software & API Development	Python, FastAPI, C++, Java, Bash, SQL, NoSQL, configurable validation engines, internal platforms
Platform Engineering	Developer tooling, CI/CD standardization, pipeline-as-code, Kubernetes, Docker, ArgoCD, artifact management
Reliability & Observability	Automated testing and validation, Dynatrace, Splunk, ELK, incident response, RCA, automated remediation
Cloud & Infrastructure	Microsoft Azure, Terraform, Ansible, Puppet, GitHub Actions, Azure DevOps, Jenkins

Work Experience

Énergir — Senior DevOps Engineer

Montréal, QC | September 2021 – Present

- Introduced AI-assisted engineering practices into the team's tooling, researching and implementing hard scope gates and human-validation checkpoints before generated code can reach production; identified and raised the absence of equivalent organization-wide safeguards to leadership.
- Owned CI/CD standardization for 12 application teams and 70+ applications across nine distinct runtimes, entering most ecosystems without prior experience and reverse-engineering each before automating delivery.
- Rebuilt Salesforce delivery with the official CLI and a validation-sandbox merge gate, reducing a recurring missed-deployment problem to one miss since launch and establishing a dependable release cadence.
- Led cloud modernization for 5–10 mission-critical services on Microsoft Azure and implemented Terraform and Ansible automation across development, staging, and production.
- Drove critical-incident RCA and implemented monitoring, observability dashboards, and automated remediation to reduce MTTR.
- Led adoption without reporting authority through recurring enablement sessions with all 12 application teams.

Key technologies: Python, Java, Microsoft Azure, Azure DevOps, GitHub Actions, Terraform, Ansible, Salesforce, Databricks, Spark, SQL

Desjardins — Senior DevOps Engineer (Contract)

Montréal, QC (Remote) | October 2025 – March 2026 — *concurrent engagement, delivered alongside the Énergir role*

- Designed and built a Python/FastAPI service that validated the configuration of roughly 250 enterprise JFrog repositories across seven package ecosystems every four hours and published results to Dynatrace.
- Extended a narrow configuration-verification assignment into a tool-agnostic validation engine with configurable test cases; the same system became the regression-safety net for the enterprise JFrog upgrade.
- Automated provisioning of on-demand Artifactory environments with Kubernetes and ArgoCD, providing administrators with self-service environments for controlled testing.

- Added artifact-governance policies and Splunk alerting for configuration changes made outside the CI/CD pipeline.

Key technologies: Python, FastAPI, JFrog Artifactory, Kubernetes, ArgoCD, GitHub Actions, Dynatrace, Splunk, Docker, Helm

Tink — DevOps Engineer

Montréal, QC | January 2018 – August 2021

- Conceived and built an internal software-inventory platform covering every installed application and version across 600+ servers, exposing CVE, end-of-life, and SSL/TLS risks to product owners who previously lacked fleet-wide visibility.
- Centralized Jenkins Pipeline as Code for 10–15 developers maintaining 80+ applications for 20+ clients, standardizing delivery across heterogeneous stacks.
- Automated operating-system patching and kernel-update reboots across the server fleet and built a self-healing response for a recurring Elasticsearch failure.
- Built the compute, caching, database, DNS, and network layers of an AWS migration target; the client cancelled the migration before production cutover.

Key technologies: Python, Bash, Jenkins, Linux, Puppet, Spacewalk, ELK Stack, AWS, Jira

SOVO Technologies — Lead Technician → Software Engineer

Montréal, QC | November 2008 – December 2017

- Wrote production software for roughly seven years before the title formalized it, including a production scheduling platform used for the department's full planning cycle.
- Built a voice-recognition evaluation system and multilingual dictionary infrastructure supporting internationalized locales.
- Owned company infrastructure, managed a technician team, and served as the final quality gate before software releases reached production.

Key technologies: C++, Java, SQL, Windows Server, Linux, Active Directory, VMware vSphere

Education

Bachelor of Automated Manufacturing Engineering — École de Technologie Supérieure (ÉTS), Montréal, QC | 2005 – 2016

DCS in Computer Science Technology — Cégep André-Laurendeau, Montréal, QC | 2002 – 2005